

SYSTEMVERILOG TESTBENCH VERIFICATION

1. Testbench Architecture

- alu_interface: Connects testbench to DUT.
- alu_transaction: Encapsulates stimulus and outputs.
- alu_generator: Constructs test scenarios.
- alu_driver: Drives data to pins.
- alu_monitor: Samples signals.
- alu_scoreboard: Compares output against Golden Reference Model and tracks coverage.
- alu_env / alu_agent: Coordinates sub-components.

2. Test Suite Executed

- Opcode Sweep
- Boundary Testing
- Carry Generation & Clear
- Zero Flag Target
- Constrained Random (1000 transactions)

3. Verification Results Summary

Metric	Result	Status
Total Transactions Tested	1,364 (Target: >1000)	PASSED
Passed Transactions	1,364 (100%)	PASSED
Failed Transactions	0 (0%)	PASSED
Functional Coverage	100.00%	PASSED
Interface Toggle Coverage	100.00%	PASSED
RTL Expression Coverage	100.00%	PASSED

4. Conclusion

The ALU design successfully passed all 1,364 functional verification checks with 100% functional coverage. The design is fully verified and ready for deployment.